

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 — DASHBOARD DEVELOPMENT TASK

Course Code	DSA0502	Course Title	Query Processing for Data Science
CO Assessed	CO5	Assessment	Tool 3 — Dashboard Development Task
Weightage	25%	Project Mode	Individual Submission
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Section / Batch	AI & DS	Faculty In-charge	Dr. B. Shyamala Gowri
Tasks Selected	1 and 2 only	Submission Date	22/08/2026

CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication.

Declaration: This submission contains the implementation, visualizations, observations, and reproducible Python code for the two selected dashboard tasks. Tasks 3–5 are intentionally not included.

1. Tools, Dataset and Methodology

Python, Pandas and Matplotlib are used throughout the work. The datasets are CSV-based and are loaded with Pandas. The dashboard requirements are addressed through summary charts, distributions, frequency visualizations, relationship plots, scatter matrices, lag analysis and autocorrelation analysis.

Software / Libraries

- Python 3.x
- Pandas — data loading, cleaning, grouping and analysis
- Matplotlib — charts and statistical visual representations
- NumPy — reproducible demonstration dataset preparation

Dataset note: The submitted analysis uses reproducible demonstration datasets generated for this dashboard task and saved as CSV files. The student-performance dataset contains attendance, internal marks, assignment marks, laboratory marks and final scores. The sales dataset contains date, product, category, quantity, revenue and region.

2. Task 1 — Student Academic Performance Dashboard

Develop an interactive dashboard using Pandas and Matplotlib to analyze student performance. The implementation covers the required bar charts, distribution plots, scatter plots, frequency charts, and a scatter matrix for relationships, patterns and outliers.

2.1 Dataset Structure

Student_ID	Attendance	Internal_Marks	Assignment_Marks	Laboratory_Marks	Final_Score
S001	84.7	57.5	68.8	92.8	70.9
S002	72.6	72.3	80.8	83.0	76.4
S003	88.8	77.8	81.2	77.3	72.4
S004	90.5	82.4	92.6	91.0	80.0
S005	64.4	83.8	87.4	84.9	88.6
S006	70.3	84.7	82.6	94.8	77.1
S007	83.2	72.2	93.6	82.6	76.4
S008	79.2	70.9	67.1	70.0	79.4

Dataset size: 60 student records and 6 variables.

2.2 Required Visualizations

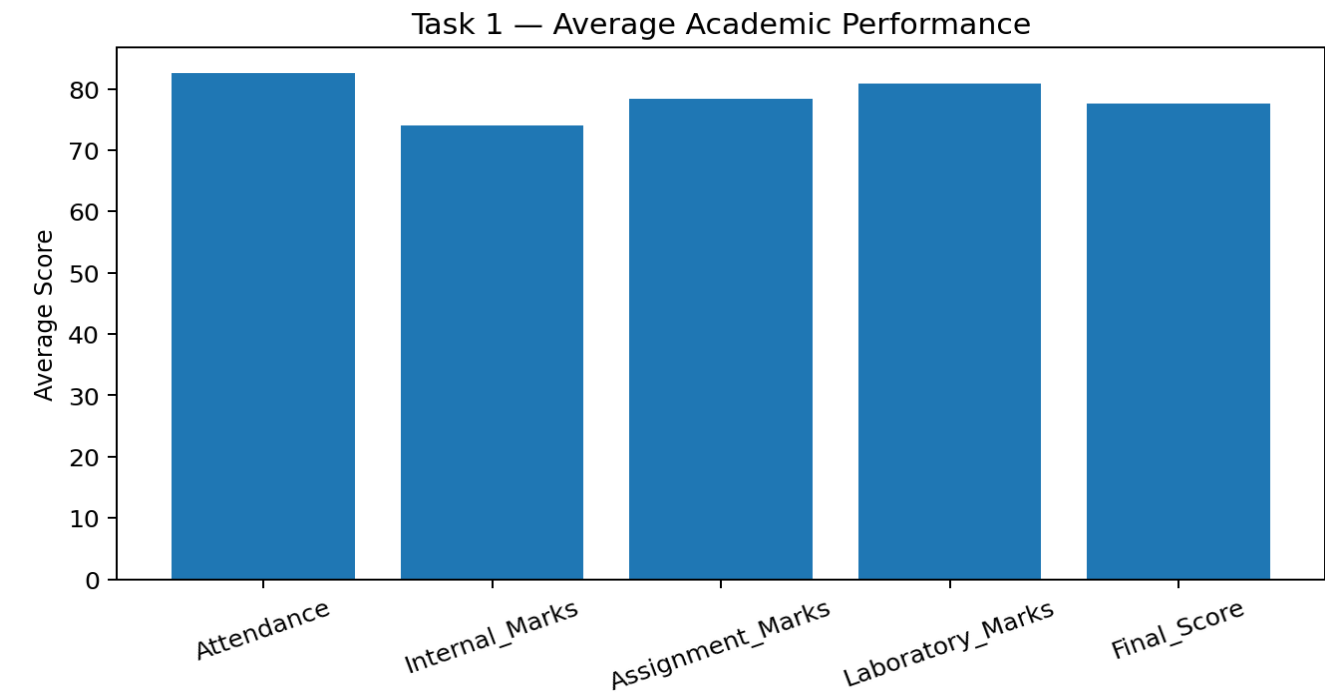


Figure 1. Bar chart showing average attendance, assessment components and final score.

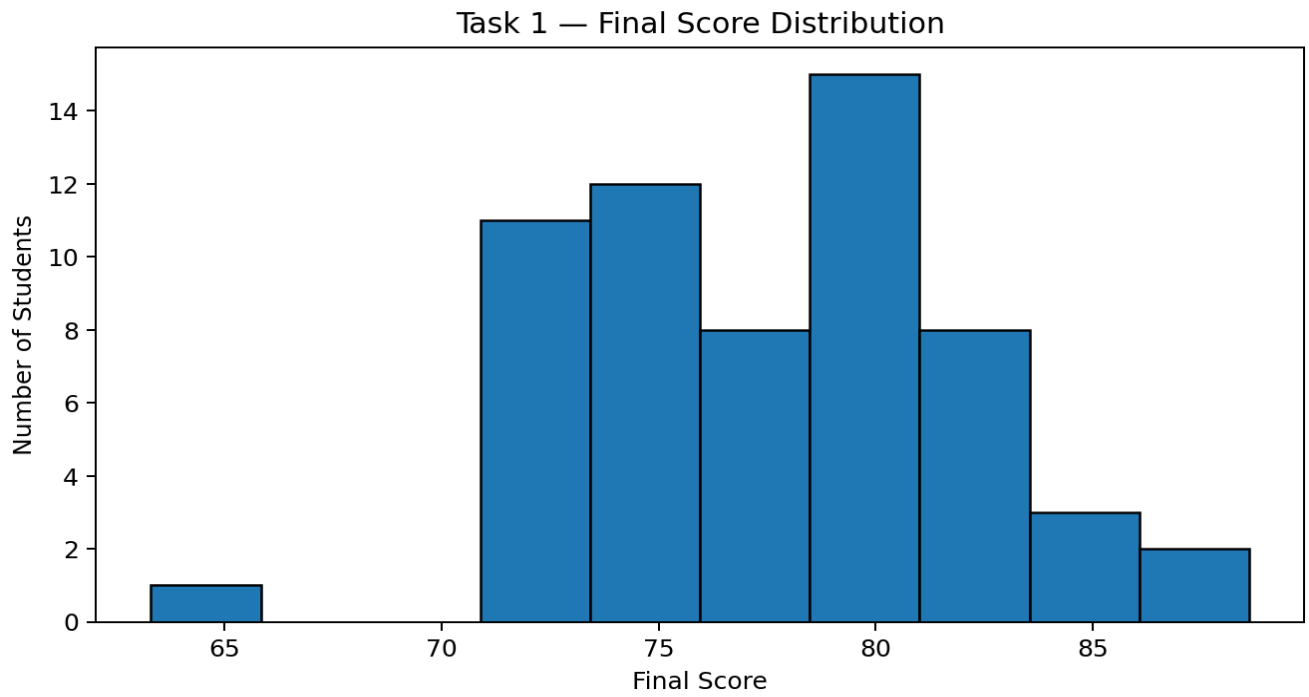


Figure 2. Distribution plot of final scores.

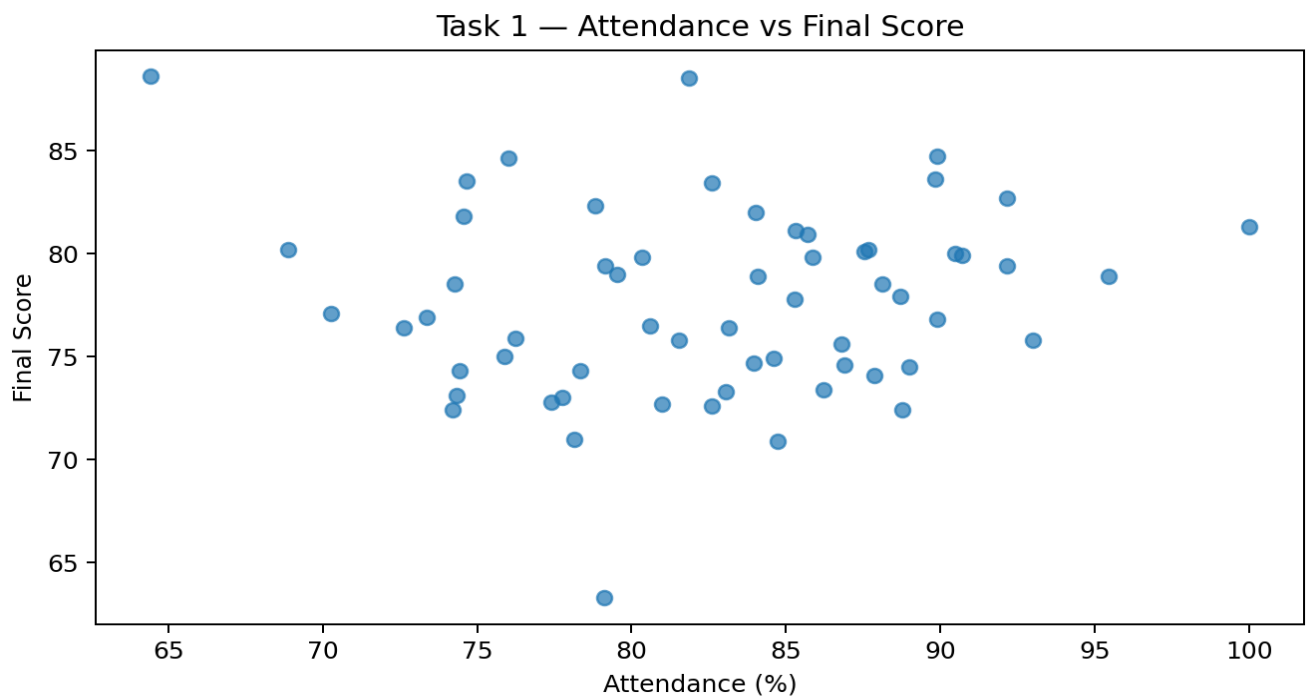


Figure 3. Scatter plot showing the relationship between attendance and final score.

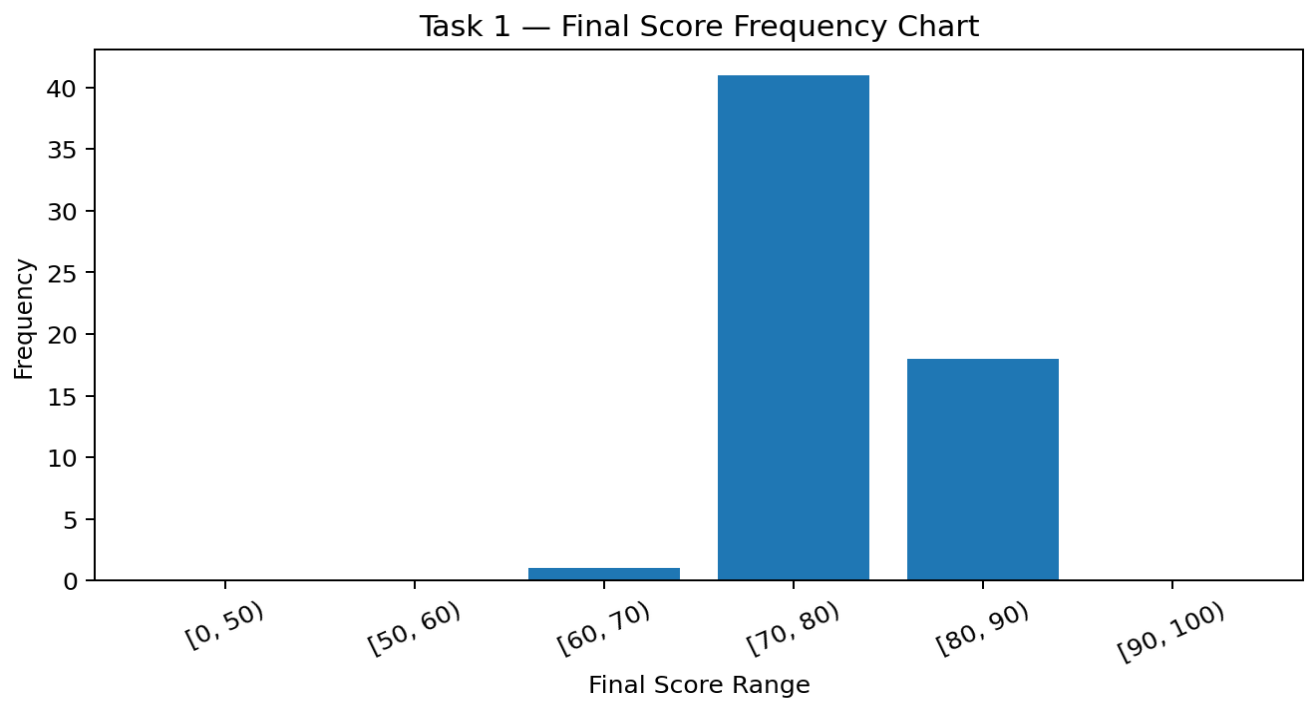


Figure 4. Frequency chart showing students across final-score ranges.

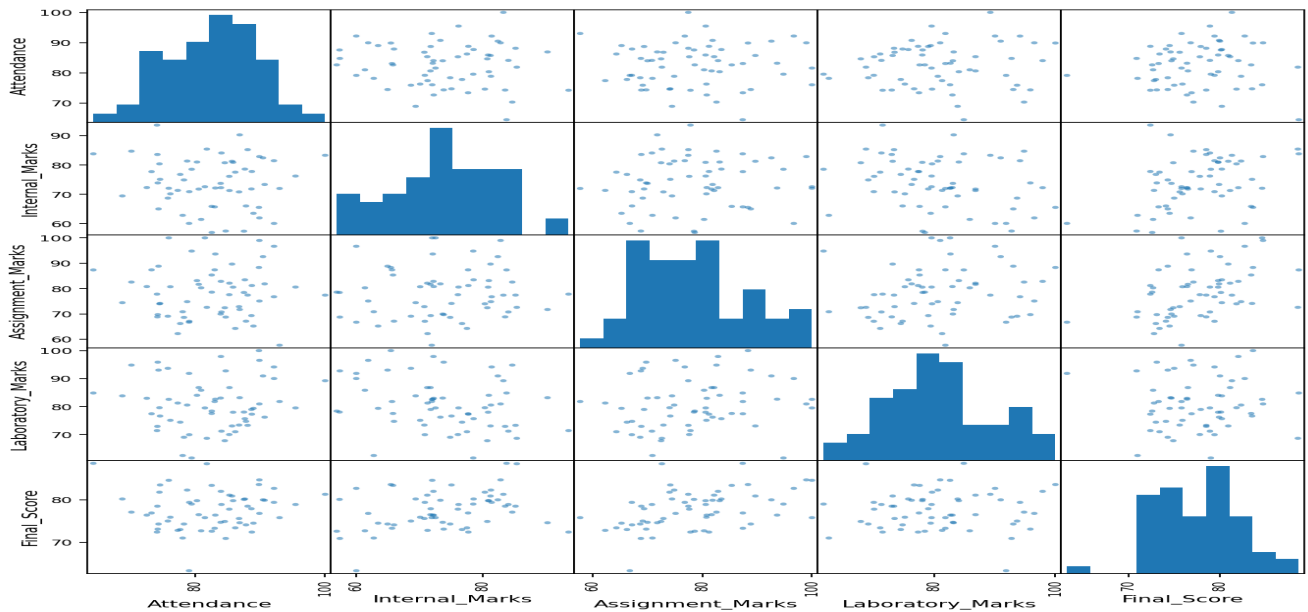


Figure 5. Scatter matrix for multivariable relationships and potential outliers.

2.3 Key Observations

- The average final score is approximately 77.6, indicating the overall performance level of the demonstration cohort.
- Attendance and final score show a positive visual relationship; students with lower attendance generally appear more frequently in the lower-score region.
- The distribution plot helps identify concentration, spread and unusually low/high final scores.
- The scatter matrix compares multiple academic variables simultaneously and can reveal clusters, correlations and potential outliers.

2.4 Implementation Code

```
import pandas as pd
import matplotlib.pyplot as plt
from pandas.plotting import scatter_matrix

df = pd.read_csv("student_performance.csv")

# Bar chart
df[["Attendance", "Internal_Marks", "Assignment_Marks",
    "Laboratory_Marks", "Final_Score"]].mean().plot(kind="bar")
plt.title("Average Academic Performance")
plt.ylabel("Average Score")
plt.tight_layout()
plt.show()

# Distribution plot
df["Final_Score"].plot(kind="hist", bins=10, edgecolor="black")
plt.title("Final Score Distribution")
plt.xlabel("Final Score")
plt.show()

# Scatter plot
plt.scatter(df["Attendance"], df["Final_Score"], alpha=0.7)
plt.xlabel("Attendance (%)")
plt.ylabel("Final Score")
plt.title("Attendance vs Final Score")
plt.show()

# Frequency chart
ranges = pd.cut(df["Final_Score"], [0, 50, 60, 70, 80, 90, 100],
    right=False).value_counts().sort_index()
ranges.plot(kind="bar")
plt.title("Final Score Frequency")
plt.xlabel("Score Range")
plt.ylabel("Frequency")
plt.show()

# Scatter matrix
scatter_matrix(
    df[["Attendance", "Internal_Marks", "Assignment_Marks",
        "Laboratory_Marks", "Final_Score"]],
    figsize=(10, 10), diagonal="hist", alpha=0.55)
```

```
)  
plt.suptitle("Academic Performance Scatter Matrix")  
plt.show()
```

3. Task 2 — Sales Performance and Trend Dashboard

Create a dashboard for an e-commerce company using sales data containing date, product, category, quantity, revenue and region. The implementation analyzes sales evolution over time, product-wise distributions and category frequencies, and includes lag and autocorrelation plots to identify time-dependent sales patterns.

3.1 Dataset Structure

Date	Product	Category	Quantity	Revenue	Region
2025-01-01	Laptop	Computers	6	4769.3	East
2025-01-01	Phone	Mobiles	10	4462.87	West
2025-01-01	Headphones	Audio	5	371.55	East
2025-01-01	Tablet	Mobiles	10	3605.0	North
2025-01-01	Smartwatch	Wearables	9	1911.91	South
2025-01-02	Laptop	Computers	9	6678.75	East
2025-01-02	Phone	Mobiles	12	5978.51	West
2025-01-02	Headphones	Audio	3	281.71	East
2025-01-02	Tablet	Mobiles	7	1883.09	North
2025-01-02	Smartwatch	Wearables	6	1160.9	West

Dataset size: 900 transaction-day-product records across 5 products, 4 categories and 4 regions.

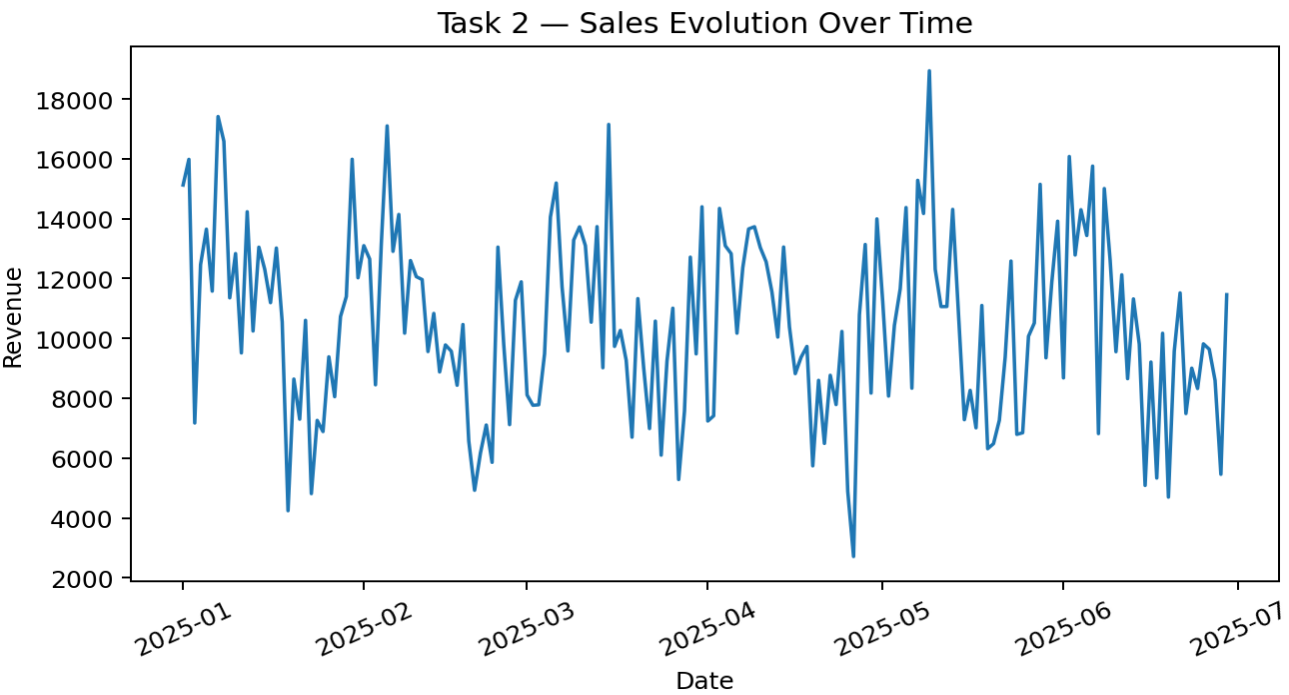


Figure 6. Daily revenue evolution across the analysis period.

Task 2 — Product-wise Revenue Distribution

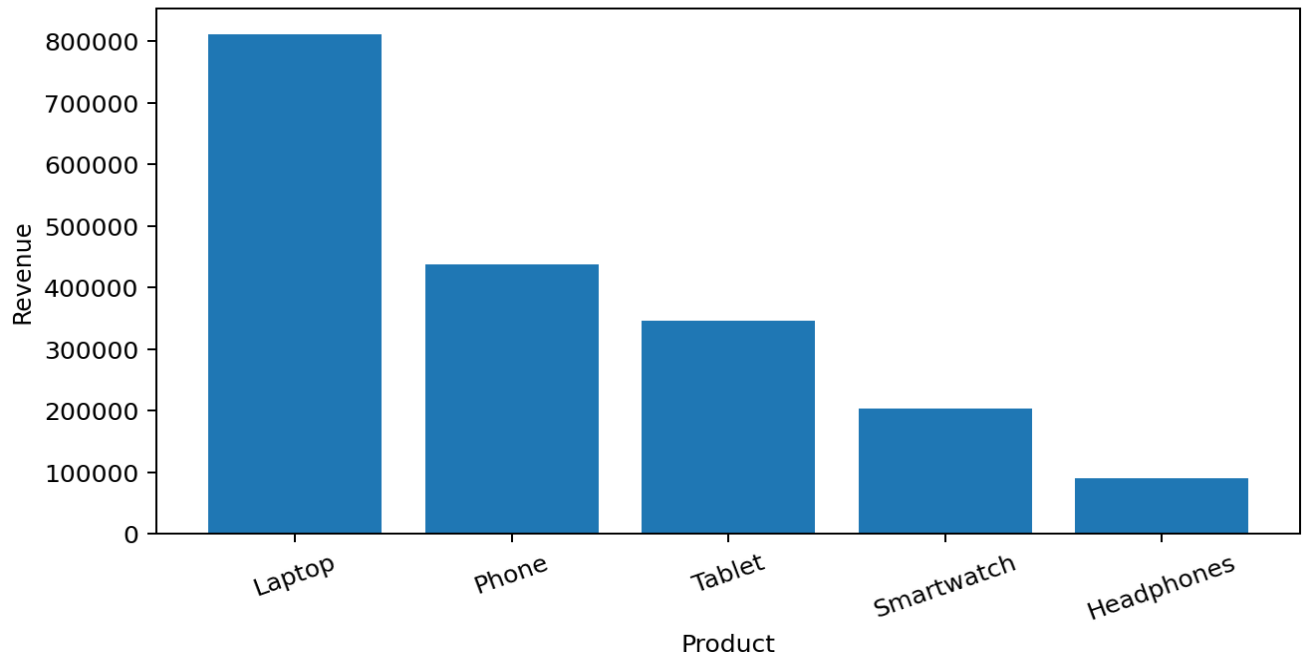


Figure 7. Product-wise revenue distribution.

Task 2 — Category Frequency

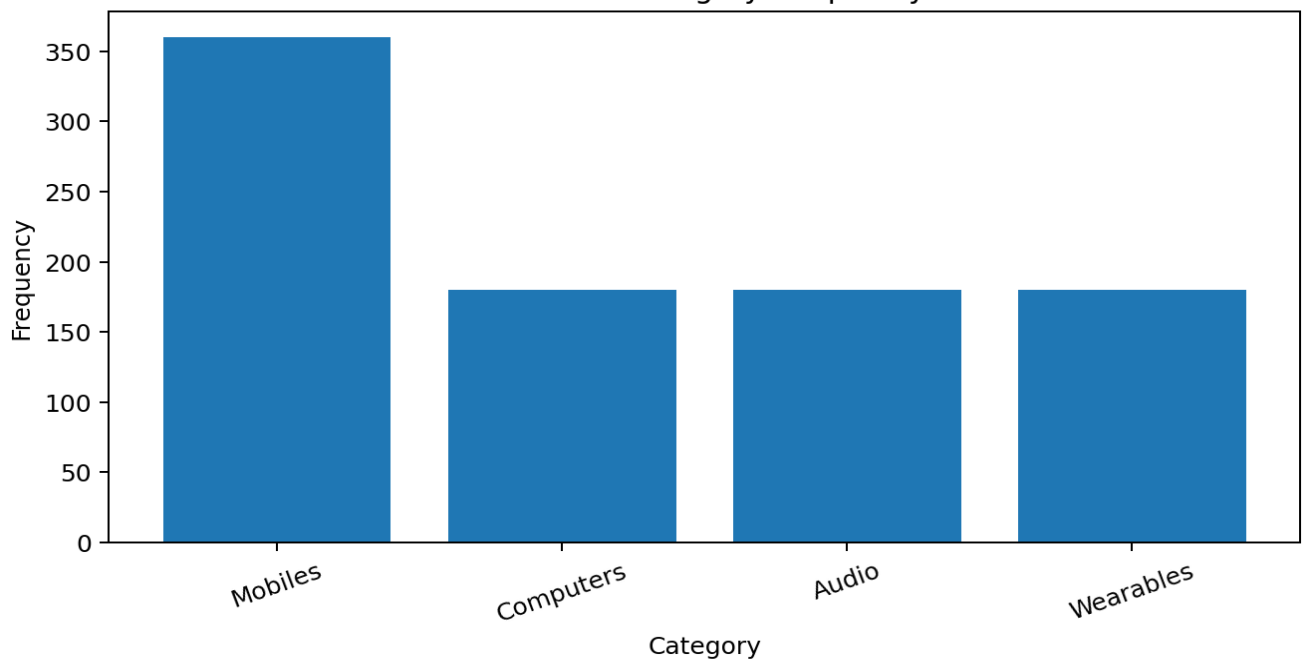


Figure 8. Category frequency chart.

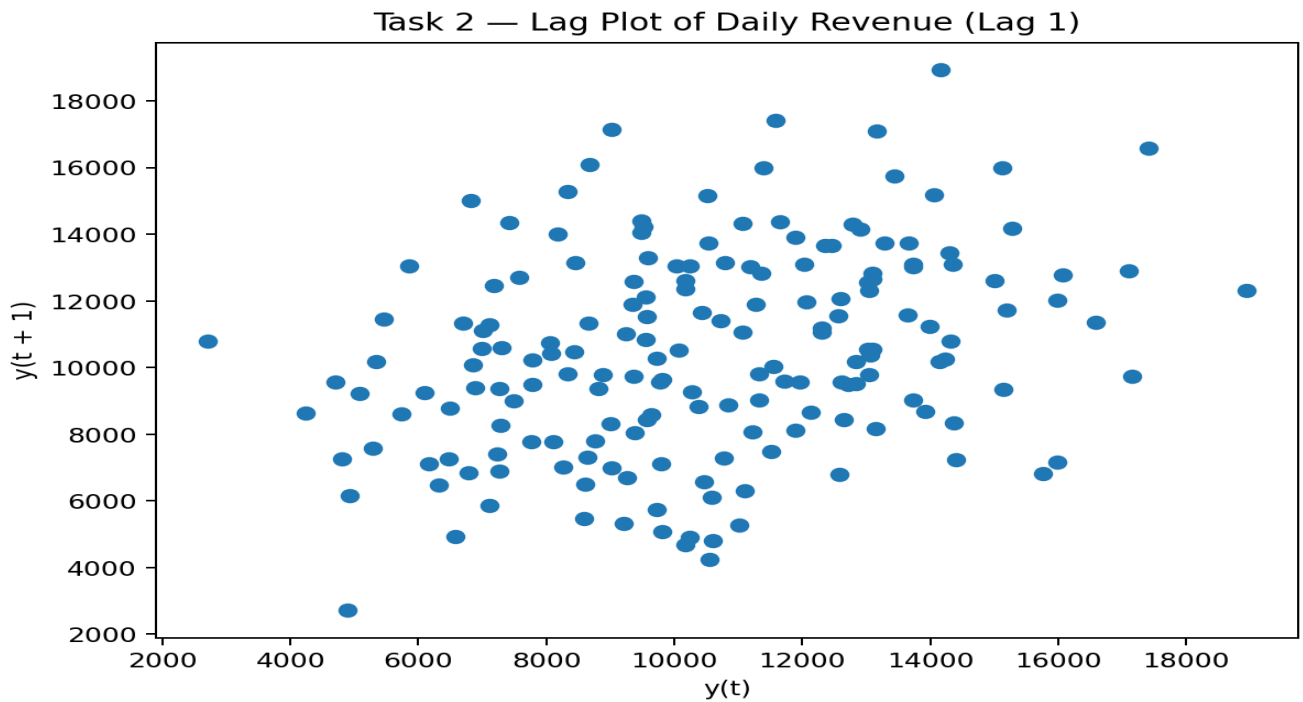


Figure 9. Lag plot showing the relationship between current and previous-day revenue.

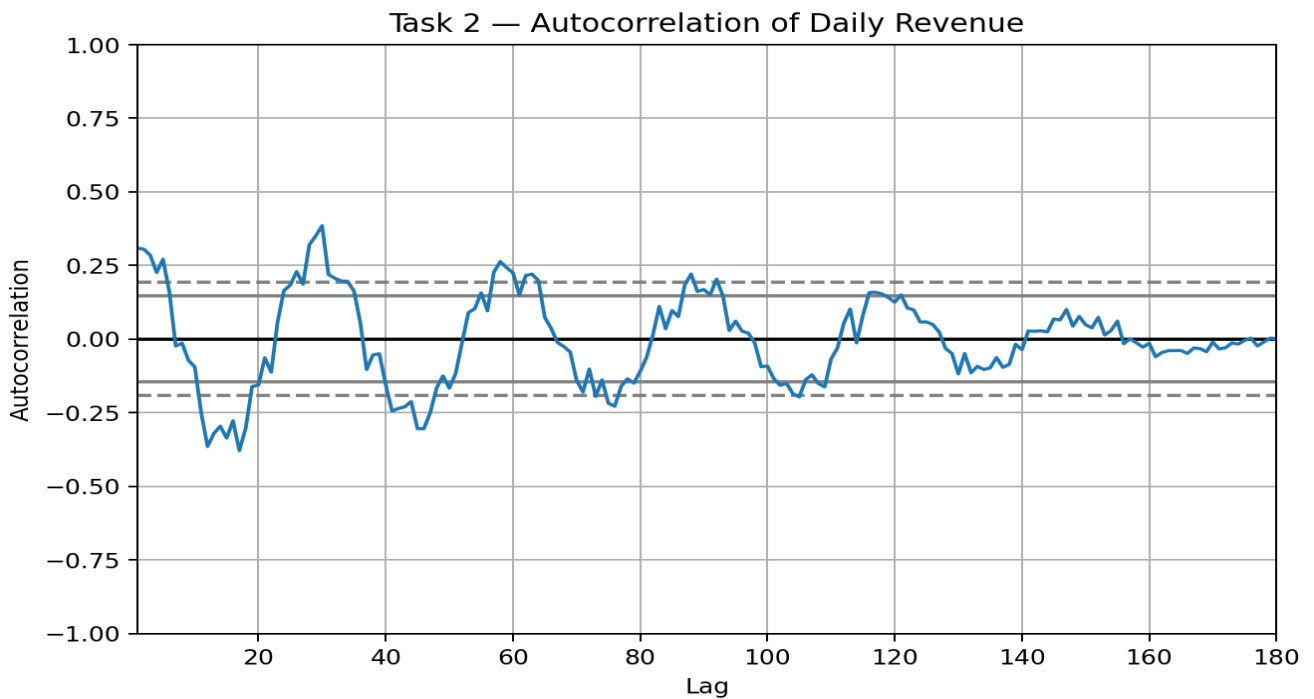


Figure 10. Autocorrelation plot for daily revenue.

3.2 Key Observations

- Total revenue in the demonstration dataset is approximately 1,888,254 currency units.
- Laptop contributes the largest product-wise revenue in this dataset, while the product chart makes the differences visible.
- The daily trend plot shows short-term fluctuations and recurring variation in sales over time.
- The lag plot compares each day's revenue with the previous day; clustering around a rising diagonal indicates positive lag dependence.
- The autocorrelation plot helps determine whether revenue observations are related across time lags, supporting time-dependent pattern analysis.

3.3 Implementation Code

```
import pandas as pd
```

```

import matplotlib.pyplot as plt
from pandas.plotting import lag_plot, autocorrelation_plot

df = pd.read_csv("sales_data.csv", parse_dates=["Date"])

# Sales evolution
daily_sales = df.groupby("Date")["Revenue"].sum()
daily_sales.plot()
plt.title("Sales Evolution Over Time")
plt.ylabel("Revenue")
plt.show()

# Product-wise distribution
df.groupby("Product")["Revenue"].sum().sort_values(
    ascending=False
).plot(kind="bar")
plt.title("Product-wise Revenue Distribution")
plt.ylabel("Revenue")
plt.show()

# Category frequency
df["Category"].value_counts().plot(kind="bar")
plt.title("Category Frequencies")
plt.ylabel("Frequency")
plt.show()

# Lag plot
lag_plot(daily_sales, lag=1)
plt.title("Lag Plot of Daily Revenue")
plt.show()

# Autocorrelation plot
autocorrelation_plot(daily_sales)
plt.title("Autocorrelation of Daily Revenue")
plt.show()

```

4. Conclusion

The two selected dashboard tasks have been implemented using Pandas and Matplotlib. Task 1 demonstrates academic-performance analysis through bar, distribution, scatter, frequency and scatter-matrix visualizations. Task 2 demonstrates e-commerce sales analysis through time-series, product, category, lag and autocorrelation visualizations. Together, the dashboards satisfy the specified visualization and relationship-analysis requirements for the selected tasks.

5. Submission Checklist

- Task 1 — Student Academic Performance Dashboard completed.
- Task 1 includes bar chart, distribution plot, scatter plot, frequency chart and scatter matrix.
- Task 2 — Sales Performance and Trend Dashboard completed.
- Task 2 includes sales trend, product-wise distribution, category frequency, lag plot and autocorrelation plot.
- Pandas and Matplotlib are used for data processing and visualization.
- Dataset structure, observations and reproducible Python implementation are included.
- Tasks 3, 4 and 5 are intentionally omitted as requested.